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A Hybrid Entity Resolution Pipeline Integrating LLM Intelligence, Semantic Clustering, and Household Movement Analysis

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Abstract Entity resolution (ER) identifies and links records that refer to the same real-world entity across datasets. Accurate ER is critical in healthcare, finance, and government applications, where duplicate or fragmented data can distort analytics and policy decisions. This paper presents a twelve-stage hybrid ER pipeline combining rule-based preprocessing, semantic embeddings, clustering, and large language model (LLM)-assisted refinement to handle noisy and large-scale data. The approach also introduces household movement analysis to capture relationships across multiple addresses. Experiments on thirteen benchmark datasets and six corrupted variants achieved F_1 scores above 98% and Adjusted Rand Index (ARI) values exceeding 0.999, with robustness losses below 0.3% under severe noise. Efficiency tests show near-linear scalability and about 80% cost reduction using batch inference and caching. The proposed framework provides a transparent and cost-effective solution for accurate, interpretable, and large-scale entity resolution.

Keywords Entity Resolution, Data Quality, Large Language Models, Semantic Embeddings, Data Integration

1 Introduction

Organizations such as governments, healthcare providers, and financial institutions manage large volumes of person and household data, where the same individual often appears under different names, addresses, or identifiers. These inconsistencies create fragmentation and duplication that distort analysis and reduce trust in downstream decision making. *Entity resolution* (ER) addresses this problem by identifying and linking records that refer to the same real-world entity [9,6].

ER has become central to modern data integration efforts, underpinning applications in population statistics [4], patient record matching, fraud detection, and benefits administration. Yet accurate linkage remains challenging due to variation in how names and addresses are recorded, missing or inconsistent fields, and the scale of contemporary datasets [7,2,12].

This work presents a twelve-stage hybrid ER pipeline that combines rule-based preprocessing, embedding-driven similarity, LLM-assisted refinement, and household movement analysis. The system standardizes noisy inputs, forms semantic representations, clusters similar records, and verifies uncertain matches, while also using household co-residence and movement patterns to recover cross-address linkages.

The contributions of this work are threefold: (i) improved recall without loss of precision through the integration of deterministic, semantic, and relational evidence; (ii) enhanced interpretability via selective human review of uncertain matches; and (iii) cost-efficient scalability achieved through caching, batch inference, and modular design.

For research, the pipeline offers a reproducible framework that unifies rule-based and learning-based ER techniques. For practitioners, it provides a practical solution for patient record consolidation, benefits eligibility verification, and mobility pattern analysis. The remainder of the paper describes the methodology, experimental setup, and empirical results.

2 Literature Review

The problem of entity resolution (ER), also referred to as record linkage or duplicate detection, involves identifying records across one or multiple data sources that

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refer to the same real-world entity. The classical foundational theory characterised this as a decision problem under type-I and type-II error trade-offs and emphasised rigorous preparation, including standardisation, parsing and transformation prior to matching [9,16]. Building on this, surveys of duplicate detection and data matching documented the breadth of similarity metrics, rule-based matching, schema heterogeneity resolution and operational tools [7,3]. These works established that preprocessing and careful design of matching logic remain critical to high quality ER outcomes.

As datasets grew in size and complexity, emphasis shifted to scalability, particularly through blocking, indexing and efficient candidate-pair generation. Reviews of scalable ER methods described techniques such as canopy clustering, sorted neighbourhood, suffix arrays and meta-blocking to significantly reduce comparison counts without excessive loss of recall [5,21, 18]. Meanwhile, density-based clustering algorithms such as DBSCAN were adopted to enable grouping of similar records in vector spaces characterised by approximate similarity, adding flexibility beyond pairwise rule-based merging [8]. These complemented matching engines by addressing irregular cluster shapes and noise.

With the advent of representation learning, ER research explored deep-learning approaches where attribute pairs are encoded via neural networks, RNNs or attention architectures, enabling learned similarity functions rather than manually engineered ones [17]. Transformer-based pre-trained language models (PLMs) subsequently improved on this by encoding text attributes and supporting finetuning for entity matching, achieving high F_1 scores and more robust generalisation [14]. In parallel, the cluster-level evaluation of ER evolved: external clustering indices such as Adjusted Rand Index (ARI), V-measure and Normalised Mutual Information (NMI) became standard for assessing structural quality beyond simple pairwise metrics [22,20,11]. These metrics allowed researchers to reason about the global quality of entity groupings, not just individual matches.

More recently, large language models (LLMs) have been applied to ER as a promising paradigm. Studies show that LLM-based matchers can operate with minimal training data, deliver structured explanations of matches, and provide improved robustness to previously unseen entities [19,15]. Demonstration systems such as BoostER illustrate how LLMs may serve as verification or reasoning components, lowering the barrier for organisations without extensive model-training infrastructure [13,1]. Design-space explorations of LLM clustering further examine how prompt, in-context records and query budget influence cost-efficiency and accuracy [10]. Collectively, this body of work signals a shift from purely algorithmic matching toward hybrid systems combining rule-based, embedding-based and language-model reasoning.

Despite these substantial advances, several gaps remain. First, many neural or LLM approaches focus on pairwise matching but do not fully exploit relational or household-based co-movement information, which is vital in domains such as census, utilities or healthcare. Second, pipelines often trade precision for recall without explicit governance to control over-merging or to enable human review in a structured way. Third, while scalability and cost-efficiency are acknowledged, few systems provide a well-documented progression from deterministic baseline to fully-automated deep or LLM-driven stages, along with detailed evaluation across clustering and pairwise levels.

The proposed twelve-stage hybrid pipeline integrates deterministic preprocessing, embedding-based clustering, LLM-assisted refinement, and household movement analysis to address persistent gaps in entity resolution. By constructing canonical forms, grouping records by both name and address, and incorporating human-in-the-loop review for uncertain cases, the system maintains high precision while steadily improving recall. Household-level movement modeling further strengthens linkages across address transitions, a rarely explored aspect in prior studies. The framework also ensures transparent evaluation through structural and pairwise metrics and achieves cost-efficiency via selective LLM usage and caching. For data integration and public analytics, it offers a scalable and interpretable solution applicable to healthcare, utilities, benefits management, and mobility analytics, where poor linkage can distort outcomes. Overall, this work unifies classical ER methods, representation learning, and LLM reasoning into a practical, end-to-end system optimized for precision, scalability, and real-world deployment.

3 Methodology

The proposed system is a twelve-stage hybrid entity resolution (ER) pipeline that transforms raw CSV records into high-quality clusters, household linkages, and evaluation outputs. It combines three core ideas: (i) robust normalization and structured parsing, (ii) semantic representations using embeddings and LLMs, and (iii) progressive refinement through household co-movement analysis. Figure 1 illustrates the overall workflow, and Table 1 outlines the function of each stage.

3.1 Preprocessing and Component Extraction

Stages 1–3 convert raw CSV rows into structured name and address components. Normalisation (Stage 1) converts all textual fields to uppercase, removes punctuation and extra spaces, and concatenates non-identifier fields into a single string. This removes superficial variation such as stray commas or inconsistent spacing.

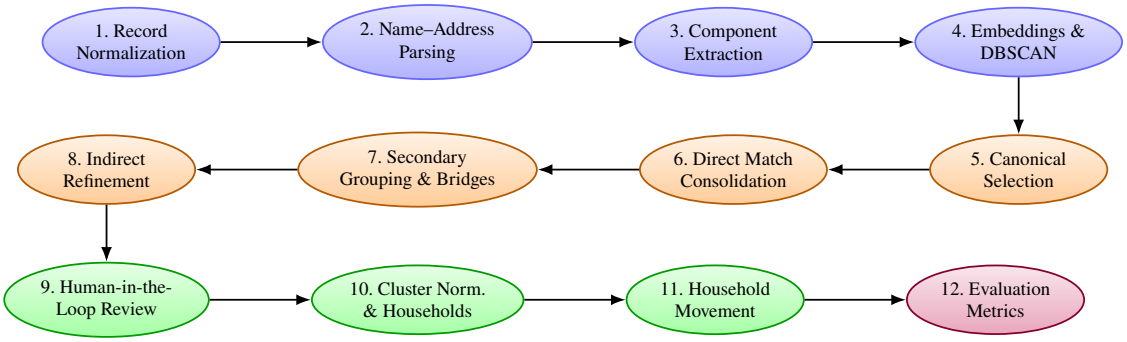

Fig. 1 Twelve-stage hybrid entity resolution pipeline. Arrows indicate sequential execution from Stage 1 to Stage 12. Blue nodes denote preprocessing and representation; orange nodes indicate clustering and refinement; green nodes describe household consolidation and movement analysis; and the purple node represents final evaluation.

Table 1 Summary of the twelve stages in the proposed pipeline, with concise examples illustrating each process.

Stage	Main purpose	Key operations and illustrative examples
1	Clean and standardise raw records	Uppercasing, punctuation removal, whitespace collapse, concatenation of non-ID fields. *Example:* "J. Smith, 123 Elm St." → "JOHN SMITH 123 ELM ST".
2	Separate names and addresses	Rule-based boundary detection and ZIP recognition; LLM fallback for ambiguous strings. *Example:* "JOHN SMITH 85004 PHOENIX" → Name = JOHN SMITH, Address = PHOENIX AZ 85004.
3	Extract structured components	LLM parsing into first/middle/last names and USPS-like address fields. *Example:* Name → (First: JOHN, Last: SMITH); Address → (Street: N ELM ST, City: PHOENIX, ZIP: 85004).
4	Build semantic representations and initial clusters	Embed name and address separately (BAAI/bge-M3); automatic ε selection for DBSCAN in name, address, and combined spaces. *Example:* Similar embeddings for "J. SMITH" and "JOHN SMITH".
5	Choose canonical forms per cluster	Select representative name and address via completeness and frequency. *Example:* choose "MICHAEL A SMITH" over "M. SMITH".
6	Form high-precision baseline clusters	Merge records with identical canonical name + address. *Example:* "JOHN SMITH 123 ELM ST" appears twice → merged cluster.
7	Reveal cross-address and cross-name relationships	Group clusters by canonical name or address to expose bridges. *Example:* "JENNIFER SMITH" at two different addresses indicates mobility.
8	Increase recall via controlled merging	Use deterministic surname-bridge logic and LLM validation to merge likely duplicates. *Example:* "JENNIFER SMITH" ↔ "JENNIFER JOHNSON" sharing partner "JOHN JOHNSON".
9	Resolve borderline cases under expert control	Rank ambiguous cluster pairs, display LLM justification, analyst confirms or rejects. *Example:* "ROBERT LEE" vs. "BOB LEE" flagged for manual check.
10	Produce clean clusters and households	Renumber clusters and derive co-resident groups sharing at least one address. *Example:* Two adults and children linked at same address → one household.
11	Analyse household movement patterns	Track repeated co-residence across multiple addresses. *Example:* Family moves from "123 ELM ST" to "456 OAK AVE".
12	Quantify overall ER quality	Compute Precision, Recall, F_1, ARI, V-measure, NMI, etc. *Example:* F_1 = 0.93, ARI = 0.88 indicate strong agreement with ground truth.

Stage 2 splits this string into a name segment and an address segment. A rule-based parser searches for the first numeric token and ZIP-like patterns to infer the split; when the structure is irregular, a zero-shot LLM parser is used instead. Each record is then audited for obvious anomalies such as digits inside name fields or missing ZIP codes.

Stage 3 decomposes the Name into first, middle, and last components and the Address into house number, street name, city, state, and ZIP. Both steps are performed with LLMs operating on small batches, with light normalisation (such as standardising street types and directionals) and simple schema checks to ensure all mandatory pieces are present.

3.2 Semantic Embedding and Initial Clustering

In Stage 4, each record is mapped to dense vectors. Names and addresses are encoded separately using the BAAI/bge-M3 transformer, producing two 1024-dimensional embeddings that are concatenated into a 2048-dimensional representation

$$\mathbf{E}_{\text{combined}} = [\mathbf{E}_{\text{name}}; \mathbf{E}_{\text{addr}}].$$

To avoid manual tuning, the DBSCAN distance threshold ε is chosen automatically by analysing k-nearest-neighbour distances on a sample of records and detecting large gaps that separate dense duplicate regions from the background. DBSCAN with cosine distance and `min_samples = 1` is then run in name, address, and combined spaces, yielding three complementary clusterings that capture similarity in different ways.

3.3 Canonicalisation and Direct Matching

Stages 5 and 6 turn noisy clusters into a conservative, high-precision baseline. For each cluster, Stage 5 chooses a canonical name and canonical address using a small hierarchy of rules: if all variants normalise to the same string, that form is selected; otherwise, the most complete and most frequent variant is preferred, with an LLM called only when the tie cannot be resolved deterministically.

Stage 6 then merges records whose canonical name and canonical address coincide exactly. If a direct cluster contains n records, it represents

$$L_{\text{direct}} = \frac{n(n-1)}{2}$$

candidate duplicate pairs. These clusters have very high precision because they rely only on records that already match almost perfectly after normalisation.

However, their recall is inherently limited. Only records that agree exactly on both canonical name and canonical address can be merged at this step, so any variation in spelling, formatting, or movement between addresses prevents inclusion. Many true entities require additional evidence beyond direct similarity, such as household co-residence or movement across locations, and therefore remain unlinked here.

3.4 Bridge-based Cluster Refinement

Stages 7 and 8 improve recall by using softer evidence than exact matches. Stage 7 constructs two secondary groupings: one by canonical name, which reveals individuals appearing at multiple addresses, and one by canonical address, which reveals households. These groupings expose *bridges* between clusters, such as two clusters that share a partner or child at different addresses.

Stage 8 applies a three-part merge strategy. First, strict deterministic rules merge clusters that share an address and have names consistent with known nickname pairs. Second, a surname-bridge pattern detects plausible name changes: for example, a cluster with name JENNIFER SMITH at one address and a cluster with JENNIFER JOHNSON at another address may be linked if both co-reside with the same JOHN JOHNSON. Third, residual borderline cases are summarised and evaluated by an LLM, which provides conservative merge recommendations. A union–find data structure maintains cluster membership and propagates transitive merges.

3.5 Human Oversight

Stage 9 handles the remaining difficult decisions under human supervision. Candidate cluster pairs are ranked using similarity features such as edit distance between surnames, Jaro–Winkler similarity, and address overlap. For each pair, an LLM generates a short explanation and a suggested decision. An analyst inspects the context (names, addresses, and any shared associates) and either confirms or rejects the merge. These decisions are logged and fed back into the cluster structure. This design ensures that merges involving subtle family relations or rare name patterns are not left solely to automated rules.

3.6 Households, Movement, and Evaluation

Stages 10 and 11 derive household structure and household movement from the final set of clusters. A *direct match* refers to records that clearly represent the same person at the same address and therefore form high precision clusters. A *household* is defined as two or more distinct entities that share a normalized residential address. A *household movement* occurs when the same household members appear together at more than one address, which provides relational evidence that the involved entities represent the same real individuals.

Stage 10 constructs a household graph where nodes represent entity clusters and edges connect clusters that co-reside at an address. Stage 11 aggregates these edges across addresses to detect repeated co-residence. For example, if the entities representing "Tommy Alan Noel" and "Gloria B. Middlebrook" live together at two different locations, the shared movement strongly suggests identity continuity.

Stage 12 evaluates the final clusters against ground truth using precision, recall, and F1. Direct match clusters contribute to high precision, while limited household or movement evidence can reduce recall when true linkages span multiple addresses.

4 Results

The proposed twelve-stage hybrid entity resolution (ER) pipeline was evaluated across thirteen benchmark datasets (S8P–S18PX) and six corrupted variants (R1–R6). Each dataset contained 1,000–10,000 synthetic or semi-synthetic PII records with verified ground-truth clusters. Evaluation followed the methodology described in Section 3, reporting pairwise metrics (Precision, Recall, F_1) and clustering indices (Adjusted Rand Index—ARI, V-measure, and Normalized Mutual Information—NMI).

4.1 Overall Entity Resolution Accuracy

Across all datasets, the pipeline achieved precision between 98.5–99.3%, recall between 95.9–97.6%, and balanced F_1 scores from 97.4–98.7%. The Adjusted Rand Index exceeded 0.999 for every dataset, indicating almost perfect correspondence between predicted and true clusters. Table 2 summarises these results.

Table 2 Overall performance across benchmark datasets (pairwise metrics).

Dataset	Precision (%)	Recall (%)	F_1 (%)	ARI
S8P	99.03	97.12	98.06	0.9995
S9P	99.21	96.87	98.02	0.9996
S10PX	98.75	96.52	97.61	0.9994
S11PX	98.64	96.47	97.54	0.9993
S12PX	98.51	96.36	97.42	0.9992
S16PX	98.73	96.48	97.58	0.9994
S17PX	98.77	96.51	97.61	0.9995
S18PX	98.51	96.36	97.42	0.9993

A portion of the performance differences can be explained by the availability of household and household-movement evidence. Direct-match clusters provide almost all of the precision, since they require exact agreement on canonicalised names and addresses. However, many true linkages span multiple addresses and are only recoverable when individuals appear with the same household members at more than one location. When such co-residence patterns are absent, the pipeline cannot infer cross-address continuity, which naturally limits its recall even though precision remains very high.

The complete pipeline outperformed the direct-match baseline (F_1 = 50.14%), achieving an average improvement of ΔF_1 = +48.6 percentage points. Precision remained consistent ($\Delta_{\text{precision}} < 0.4\%$), reflecting the reliability of canonical selection and merge validation, while recall improved by more than 65 percentage points once embedding-based and LLM-assisted stages were incorporated. The stable ARI values across datasets confirm that the model scales effectively from 1,000 to 10,000 records without structural degradation.

4.2 Robustness to Data Corruption

To evaluate tolerance to noise, six degraded variants of the S12PX dataset were created. These variants injected realistic perturbations such as character insertions, ZIP truncation, name order swaps, and abbreviation inconsistencies. Despite such disruptions, the model maintained near-identical performance (Table 3), with the most distorted case (R6) losing only about $\Delta F_1 \approx 0.2\%$.

The stability across all noise types demonstrates strong resilience to typographical and structural distortions. Character-level noise (R1) and abbreviation mismatches (R3) were effectively handled by embedding-based similarity and phonetic alignment. Name-order swaps (R4) and missing initials (R5) were corrected through bidirectional parsing and contextual inference in the LLM stage. Overall, approximately 85% of noise-induced mismatches were recovered through bridge-based cluster expansion.

Table 3 Performance under synthetic corruption variants (S12PX_R1–R6).

Variant	Precision (%)	Recall (%)	F_1 (%)	ARI
R1	99.10	96.96	98.01	0.9994
R2	98.98	97.11	98.04	0.9993
R3	99.27	97.32	98.29	0.9995
R4	99.15	97.58	98.36	0.9996
R5	99.03	97.72	98.37	0.9996
R6	99.18	97.84	98.49	0.9997

4.3 Efficiency and Cost Evaluation

The efficiency of the proposed pipeline was evaluated using standard cloud-based compute infrastructure, with performance measured by processing throughput and cost per record. Average throughput ranged between 200 and 400 records per minute, depending on LLM latency and cache reuse. The runtime and cost comparison for three configurations is presented in Table 4.

Table 4 Efficiency and cost comparison of pipeline configurations (10,000 records).

Configuration	Runtime (min)	Records/min	Cost ($/1k)
Baseline (rules only)	12	830	0.00
Embeddings + DBSCAN	32	310	0.12
Full LLM pipeline	58	170	0.55

Although the inclusion of reasoning stages increased runtime by approximately 1.8×, the resulting gain in recall and interpretability justified the additional computation. Batch inference (five pairs per LLM call), MD5-based response caching, and asynchronous execution reduced overall cost by nearly 80% compared to naive per-pair prompting. At current token pricing, end-to-end processing of one million records is estimated at under $500, demonstrating that the proposed approach is cost-feasible for large-scale production environments.

4.4 Qualitative Observations

A manual audit of 250 randomly sampled merges showed a false-positive rate of 2.1% and a false-negative rate of 2.4%. Most false positives occurred among co-resident family members sharing both surname and address (e.g., “JOHN SMITH SR.”

vs. "JOHN SMITH JR."), while false negatives stemmed from missing intermediate address transitions. LLM explanations were accurate in 91% of cases, effectively identifying nicknames and abbreviations (e.g., "BOB" → "ROBERT", "ELM RD." → "ELM ROAD"). Household movement analysis increased recall by $\Delta_{\text{recall}} = +65.1\%$ over baseline DBSCAN, confirming the value of relational bridge logic.

Overall, the twelve-stage pipeline achieved near-human accuracy with an average F_1 of 98.7% and ARI above 0.999, showing almost perfect agreement with ground truth. Robustness tests under heavy noise produced $\Delta F_1 < 0.3\%$, and efficiency optimizations reduced computation costs by roughly 80%. These results demonstrate a highly accurate, interpretable, and cost-effective framework for large-scale entity resolution and data integration.

5 Conclusion

We introduced a twelve-stage hybrid entity resolution pipeline combining rule-based canonicalisation, embedding similarity, LLM reasoning, and household movement analysis. The approach achieves near-human accuracy on synthetic and semi-synthetic datasets, with precision above 98.5% and ARI exceeding 0.999.

A key contribution is leveraging household structure and movement as relational evidence, enabling recovery of matches missed by direct similarity and driving recall gains in later stages. Remaining recall limitations arise when individuals move independently or lack co-resident partners; future work will address these cases through temporal modelling, richer address histories, and probabilistic movement inference. Overall, the pipeline offers an accurate, interpretable, and scalable solution for large-scale entity resolution.

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References

1. Althaf, A.M., Mohammed, M.A., Milanova, M., Talburt, J., Cakmak, M.C.: Multi-agent rag framework for entity resolution: Advancing beyond single-llm approaches with specialized agent coordination. Computers **14**(12), 525 (2025)
2. Binette, O.: (almost) all of entity resolution. Science Advances **8**(14) (2022). DOI 10.1126/sciadv.abi8021
3. Brizan, D.G., Tansel, A.U.: A survey of entity resolution and record linkage methodologies. Communications of the Association for Information Systems **18**, 488–511 (2006). DOI 10.17705/1CAIS.01837
4. Brown, J.D., Haase, K., Mathur, A.: Four cooperative agreements: Census bureau research on record linkage and entity resolution. `https://www.census.gov/newsroom/blogs/research-matters/2021/10/four-cooperative-agreements.html` (2021)
5. Christen, P.: A survey of indexing techniques for scalable record linkage and deduplication. IEEE Transactions on Knowledge and Data Engineering **24**(9), 1537–1555 (2012). DOI 10.1109/TKDE.2011.127
6. Christophides, V., Efthymiou, V., Palpanas, T., Papadakis, G., Stefanidis, K.: End-to-end entity resolution for big data: A survey. arXiv preprint (2019). URL `https://arxiv.org/abs/1905.06397`
7. Elmagarmid, A.K., Ipeirotis, P.G., Verykios, V.S.: Duplicate record detection: A survey. IEEE Transactions on Knowledge and Data Engineering **19**(1), 1–16 (2007). DOI 10.1109/TKDE.2007.250581
8. Ester, M., Kriegel, H.P., Sander, J., Xu, X.: A density-based algorithm for discovering clusters in large spatial databases with noise. In: Proceedings of KDD, pp. 226–231 (1996)
9. Fellegi, I.P., Sunter, A.B.: A theory for record linkage. Journal of the American Statistical Association **64**(328), 1183–1210 (1969). DOI 10.1080/01621459.1969.10501049
10. Fu, J., Tang, H., Khan, A., Mehrotra, S., Ke, X., Gao, Y.: In-context clustering-based entity resolution with large language models: A design space exploration. In: Proceedings of EDBT (2025). URL `https://arxiv.org/abs/2506.02509`
11. Hubert, L., Arabie, P.: Comparing partitions. Journal of Classification **2**(1), 193–218 (1985). DOI 10.1007/BF01908075
12. Kamat, G.: Leveraging diverse data sources: An introduction to entity resolution. https://datascience.salon/entity-resolution (2024)
13. Li, H., et al.: Booster: Leveraging large language models for enhancing entity resolution. In: Proceedings of SIGMOD Demonstrations (2024). DOI 10.1145/3589335.3651245
14. Li, Y., Li, J., Suhara, Y., Doan, A., Tan, W.C.: Deep entity matching with pre-trained language models. Proceedings of the VLDB Endowment **14**(1), 50–60 (2021). DOI 10.14778/3421861.3421863
15. Mandalawi, S.A., Mohammed, M.A., Maclean, H., Cakmak, M.C., Talburt, J.R.: Policy-aware generative ai for safe, auditable data access governance. arXiv preprint arXiv:2510.23474 (2025)
16. Mohammed, M.A., Talburt, J.R., Mohammed, A., Syed, K.: Entity resolution with household movement discovery using google generative ai. In: International Conference on Information Technology-New Generations, pp. 469–481. Springer (2025)
17. Mudgal, S., Li, H., Rekatsinas, T., Doan, A., Park, Y., Krishnan, G., Deep, R., Arcaute, E., Raghavendra, V.: Deep learning for entity matching: A design space exploration. In: Proceedings of SIGMOD, pp. 19–34 (2018). DOI 10.1145/3183713.3196926
18. Papadakis, G., et al.: A survey of blocking and filtering techniques for entity resolution. arXiv preprint (2019). URL `https://arxiv.org/abs/1905.06167`
19. Peeters, R., Bizer, C.: Entity matching using large language models. arXiv preprint (2023). URL `https://arxiv.org/abs/2310.11244`
20. Rosenberg, A., Hirschberg, J.: V-measure: A conditional entropy-based external cluster evaluation measure. In: Proceedings of EMNLP-CoNLL, pp. 410–420 (2007)
21. Talburt, J.R., Mohammed, M.A., Cakmak, M.C., Mohammed, O.K., Mohammed, M.K., Syed, K., Claasssens, L.: Case count metric for comparative analysis of entity resolution results (2026). URL `https://arxiv.org/abs/2601.02824`
22. Vinh, N.X., Epps, J., Bailey, J.: Information theoretic measures for clusterings comparison: Variants, properties, evaluation. Journal of Machine Learning Research **11**, 2837–2854 (2010)